

LEARN · FREE · START WITH NO KNOWLEDGE, FINISH WITH A LOT

Learn NOSHASHI, and the XRP Ledger underneath it

A complete course in plain language. Part one teaches the XRP Ledger from the ground up: how it agrees on the truth, how accounts and tokens work, what issuers can do, how trading and payments really behave, and the compliance rules around it. Part two teaches every screen of the NOSHASHI desktop app, step by step, with real button names and real readings. Part three puts it together in everyday workflows. Then test yourself, and keep the word list.

PART 1 · 12 LESSONS · ~60 MIN

The XRP Ledger

Consensus, accounts, keys, tokens, issuer powers, the DEX, AMMs, payments, NFTs, amendments, compliance.

PART 2 · 7 LESSONS · ~50 MIN

The NOSHASHI app

How it works, setup, verdicts and receipts, and every screen explained step by step.

PART 3 · 8 WORKFLOWS

Putting it to work

Real tasks from start to finish, from vetting a token to handing an auditor a signed export.

PART 4 AND 5

Knowledge check and word list

55 questions in five rounds, including vocabulary and how it helps institutions, and 110 terms in one searchable list.

START HERE · VIDEOS

Watch first: 7 short narrated videos

7 VIDEOS · ABOUT 18 MINUTES IN ALL · CAPTIONS ON

 SUPPORT

Each video covers one idea from the course, with a voiceover and captions. Every figure in them is a real reading already quoted in the lessons, with its ledger index and date. Where a screen needs a live ledger connection, the video explains the screen rather than showing invented results.

1. 1 · What NOSHASHI does (2:32): noshashi.app/learn/videos/01-what-noshashi-does.mp4
2. 2 · XRP Ledger basics (2:35): noshashi.app/learn/videos/02-xrpl-basics.mp4
3. 3 · What an issuer can still do to your tokens (2:49): noshashi.app/learn/videos/03-issuer-powers.mp4
4. 4 · Listed depth versus funded depth (2:28): noshashi.app/learn/videos/04-funded-depth.mp4
5. 5 · Payments: what really arrived (1:49): noshashi.app/learn/videos/05-payments.mp4
6. 6 · How NOSHASHI helps institutions and enterprises (2:26):
noshashi.app/learn/videos/06-for-institutions.mp4
7. 7 · The app, screen by screen (3:22): noshashi.app/learn/videos/07-app-tour.mp4

More videos from the XRPL community

Not made by NOSHASHI. These open on YouTube, the XRPL Learning Portal and XRPL Commons, and are the best free places to go deeper on the ledger itself.

[Intro to the XRP Ledger](#)

YouTube playlist · a video series on the ledger's key concepts (goes with lessons 1 to 4)

[Intro to the XRPL](#)

XRPL Learning Portal · free course: the ledger's history, how it works, key features

[Order books on the XRPL](#)

XRPL Learning Portal · part of the free "Deep dive into XRPL DeFi" course (goes with lessons 7 and 8)

[Introduction to the XRP Ledger](#)

XRPL Commons Academy · accounts, tokens, the DEX and consensus

[XRPL Commons](#)

YouTube channel · short tutorials, each on one transaction type

[XRP Ledger Apex 2024](#)

YouTube playlist · talks from the XRPL developer conference

The official reference pages behind the most important lessons: [clawback](#), [the DEX](#), [AMMs](#), [partial payments](#) and [permissioned domains](#).

PART 1 · THE XRP LEDGER

1 • How the ledger agrees on the truth

ABOUT 6 MINUTES

The **XRP Ledger (XRPL)** is a public database, run since 2012, that keeps one shared record of who owns what. Thousands of servers around the world keep identical copies. Anyone can read it. Nobody, including its creators, can quietly change it.

Ledgers, one after another

The record moves forward in steps. Every 3 to 5 seconds the network produces a new **ledger**: a complete snapshot of every account and balance, plus the transactions that changed things since the last one. Each ledger has a number, its **ledger index**, which only ever goes up. At the time of writing it is past 107 million.

Ledger 107,207,970 → 3–5 s → Ledger 107,207,971 → 3–5 s → Ledger 107,207,972

Consensus: how strangers agree

There is no mining. Instead, special servers called **validators** compare the transactions they have seen and vote. Each server trusts a list of validators it believes will not collude, called its **UNL** (Unique Node List). When at least **80%** of trusted validators agree on the same set of transactions, that ledger becomes a **validated ledger**. From that moment it is final: it cannot be reversed or edited.

Ledgers exist in three states:

- **Open / current**: the ledger still collecting transactions. Its contents can change.
- **Closed**: no more transactions, but not yet agreed.
- **Validated**: agreed and final. This is the only kind NOSHASHI reads.

THINK OF IT LIKE

A notebook that thousands of accountants keep at once. Every few seconds they compare their newest page. When 80% of the ones you trust have written the same page, it is glued into the book forever and given a page number.

Why this matters for NOSHASHI

Because validated ledgers never change, a reading taken "at ledger 107,207,971" can be checked by anyone, at any time, by asking any server for that same ledger. Every NOSHASHI result carries its ledger index for exactly this reason.

KEY IDEA Validated means final. A ledger index is a timestamp nobody can argue with.

2 • Accounts, keys and reserves

ABOUT 6 MINUTES

Addresses

An **account** is a place on the ledger that can hold XRP and tokens. It is identified by an **address** that starts with **r** and is 25 to 35 characters long, for example Bitstamp's issuing account **rvYAfWj5gh67oV6fW32ZzP3Aw4Eubs59B**. There is also a newer **X-address** format that packs an address and a destination tag into one string.

Keys

Control of an account comes from a secret key. The **secret** (or seed, usually starting with **s**) generates a key pair: the private key signs transactions, the public key lets everyone check the signature. Anyone with the secret controls the account completely, so it must never be shared. NOSHASHI never asks for one: it only reads public data.

Reserves

To stop people filling the shared ledger with junk, every account must keep some XRP locked as a **reserve**: a base reserve to exist (currently 1 XRP) plus an **owner reserve** for each object it owns, such as a trust line or an offer (currently 0.2 XRP each). Validators

can vote to change these amounts. Reserved XRP cannot be spent, which is why NOSHASHI distinguishes a balance from what is actually **spendable**.

What an account records

FIELD	WHAT IT HOLDS
Balance	Its XRP, in drops (1 XRP = 1,000,000 drops).
Sequence	A counter that goes up with each transaction; starts at the ledger index when the account was created, not at zero.
Flags	On/off settings, including the issuer powers in lesson 5.
OwnerCount	How many objects it owns, which sets its owner reserve.
RegularKey	An optional second key (lesson 6).
Domain	An optional web domain the owner claims; can be verified against a file on that domain.
TransferRate	For issuers, the fee charged when holders transfer its tokens (lesson 5).

A COMMON MISREADING

An account's Sequence is not how many transactions it has made. It starts at the ledger index of the account's creation. NOSHASHI's Provenance screen corrects for this when it estimates account age.

KEY IDEA Whoever holds the key owns the account. Balance minus reserve is what can actually move.

3 • Transactions and what their results mean

ABOUT 5 MINUTES

Every change to the ledger is a **transaction**: a signed instruction such as Payment, OfferCreate (place a trade), TrustSet (open a trust line) or AccountSet (change a setting). Each one pays a tiny fee in XRP, called the **transaction cost**, which is destroyed rather than paid to anyone. When the network is busy the cost rises; NOSHASHI shows this as **fee pressure**.

Life of a transaction

- 1 The owner signs it with their key and submits it to a server.
- 2 The server gives a **tentative** result. This is not final.
- 3 Validators include it in a ledger that becomes validated.
- 4 Its **final result** is now fixed in the transaction's metadata.

Result codes

PREFIX	MEANING	EXAMPLE
tes	Success. The transaction did what it said, within its rules.	tesSUCCESS
tec	Failed, but still recorded and the fee was charged.	tecPATH_DRY (no way to deliver)
tef / tel / tem / ter	Not applied: malformed, rejected locally, or to retry. No fee, not in the ledger.	temBAD_AMOUNT

SUCCESS DOES NOT MEAN "ALL OF IT ARRIVED"

A payment can show **tesSUCCESS** and deliver a tiny fraction of its stated amount. Lesson 9 explains how and what to check instead.

KEY IDEA Only a result in a validated ledger is final. Read the metadata, not the request.

4 • Tokens, issuers and trust lines

ABOUT 6 MINUTES

Besides XRP, the ledger carries **tokens** (also called issued currencies or IOUs): digital dollars and euros, gold, tokenized treasury bills, shares. Every token is created by an **issuer**, an account that promises the token is backed by something. Stablecoins such as Ripple's RLUSD, Braza's BBRL, Schuman's EURØP and Novatti's AUDD are all issued this way on the XRPL.

Trust lines

To hold a token you open a **trust line** to its issuer with a TrustSet transaction. A trust line is a two-way link between you and the issuer that records a **limit** (the most you are willing to hold) and your **balance**. Each trust line you own counts toward your reserve.

Your account – trust line (limit 10,000 USD, balance 2,500) – Issuer rvYAf...

Currency codes do not prove anything

A token is identified by two things together: its **currency code** (a 3-character code like **USD**, or a 40-character hex code for longer names) **and** its issuer address. Anyone can issue a token called USD. The code is a label; the issuer is the identity. Scammers exploit this by sending look-alike tokens to people's accounts.

How much is out there

An issuer's **obligations** are the total of its tokens held by others, readable with the **gateway_balances** call. That is the real circulating supply per currency, and it is what NOSHASHI reports as outstanding.

THINK OF IT LIKE

A gift card. It is only worth something because the shop that issued it promises to honour it, and the shop printed the rules on the back. A card with the same name from a different shop is a different card.

KEY IDEA Ask "who issued this?" before "what is it called?". The issuer address is the token's identity.

5 • What an issuer can still do to your tokens

ABOUT 8 MINUTES · THE MOST IMPORTANT LESSON

Issuers can turn on powers, stored as **flags** on their account and settings on each trust line. They are public, but almost no wallet shows them. They decide whether a token is really yours.

POWER	LEDGER NAME	WHAT IT MEANS FOR A HOLDER
Individual freeze	<code>lsfLowFreeze / lsfHighFreeze</code>	The issuer freezes one holder's trust line. That holder can only send the token back to the issuer.
Global freeze	<code>lsfGlobalFreeze</code>	Every holder is frozen at once; the token stops moving for everyone except back to the issuer.
Deep freeze	Deep Freeze amendment (XLS-77)	A frozen holder can neither send nor receive the token, closing gaps the older freeze left.
No Freeze	<code>lsfNoFreeze</code>	The issuer permanently gives up freezing. It can never be switched back off.
Require Auth	<code>lsfRequireAuth</code>	You need the issuer's approval before you can hold the token. Used by regulated issuers.
Clawback	<code>lsfAllowTrustLineClawback</code>	The issuer can take tokens back out of your account. It can only be switched on before the issuer has any trust lines, so it is permanent for that token.
Transfer fee	<code>TransferRate</code>	A percentage (0% to 100%) charged each time holders send the token to each other. The issuer can change it.

Real readings

These are real certificates read from validated mainnet on 24 September 2026. You can re-run any of them for free on the [certificate page](#).

ISSUER	LEDGER	NO FREEZE	SINGLE KEY SIGNS	TRANSFER FEE	VERDICT
Bitstamp rvYAf...s59B	107,207,971	Not set	Yes (regular key)	0.15%	NO-GO
GateHub rhub8...F3wq	107,207,973	Not set	Yes (regular key)	0.20%	NO-GO
Sologenic rsoLo...rLZz	107,207,975	Set	No, blackholed	0.01%	HOLD

How to read this: the NO-GO verdicts come from a **blocking** check, “no single key controls the issuer”, failing. That is not an accusation. Keeping freeze rights is normal and often required for a regulated issuer. The certificate simply states what the issuer can still do, so a holder decides knowingly. Sologenic scores HOLD, not GO, only because it charges a transfer fee (a warning) and supply concentration was not measured.

KEY IDEA A token can be yours on paper and still be frozen or taken back. Read the issuer’s powers before you hold it.

6 • Who really controls an account

ABOUT 6 MINUTES

An account can be signed for in three ways, and more than one can be active at once:

- **Master key**: the original key pair. It can be disabled with `lsfDisableMaster`, but only if another way to sign exists.
- **Regular key**: a second key the account nominates. Useful for keeping the master key offline; it can be changed or removed.
- **Signer list** (multi-signature): up to 32 signers, each with a **weight**. A transaction is valid when the weights of those who signed add up to the **quorum**.

KEY IDEA Count weights, not people, and check whether a master key can skip the whole process.

When a new offer matches an existing one, they **cross** and trade. Token-to-token trades can be **auto-bridged** through XRP when that gives a better price.

Listed depth versus funded depth

Here is the catch that most tools miss. An offer stays on the book even after its owner no longer has the funds to fill it. It is still listed, but it cannot trade. These are **unfunded offers**. So:

- **Listed depth** = everything shown on the book.
- **Funded depth** = only what owners can actually deliver.

The ledger reports this itself. When you ask a server for a book with `book_offers`, an offer whose owner cannot cover it comes back with an extra field, `taker_gets_funded`, showing how much they really have. Adding up the listed amounts overstates the market.

WORKED EXAMPLE

Three offers to sell a token, each listing 10,000. The first owner holds 10,000, the second 500, the third 0. Listed depth: 30,000. Funded depth: 10,500. If you tried to sell 20,000 into this book, most of it would not fill.

A real measurement

Six XRPL pairs, measured with the same method on 8 September 2026 at ledger 106,850,266. "Ratio" is listed depth divided by funded depth within 10% of the mid price.

PAIR	LISTED	FUNDED IN BAND	UNFUNDED	RATIO
USD · Bitstamp	567,974	565,590	0.1%	1.0×
USD · GateHub	248,521	234,690	0.1%	1.1×
EUR · GateHub	249,570	130,588	0.0%	1.9×
CNY (rKiCet...)	932,823	135,270	27.4%	6.9×
SOL0 · Sologenic	17,573,534	6,906	94.7%	2,545×
BTC · Bitstamp	115	0	90.3%	4,841×

Median 4.4×. Deep stablecoin books are close to honest; the gap is concentrated in long-tail assets.

THREE HONEST DISCLOSURES

The 10% band is a parameter we chose; a wider band gives a smaller gap. AMM pools are left out, so where a pool exists the real exit is better. And `book_offers` returns at most 60 price levels. All three push the gap upward. Books change every few seconds: re-measure before you quote any figure.

KEY IDEA Listed depth is an advert. Funded depth is what you can actually get out at.

8 • AMM pools

ABOUT 5 MINUTES

Since the AMM amendment ([XLS-30](#)), the XRPL also has **automated market makers**: pools that hold two assets, for example XRP and a stablecoin, and trade against anyone at a price set by the ratio of the two balances. The DEX routes trades through both order books and pools, whichever is better.

- **Liquidity providers** deposit both assets and receive **LP tokens**, a share of the pool.
- Every swap pays a **trading fee** (up to 1%) that goes to the pool. LP token holders **vote** on the fee, weighted by how many LP tokens they hold.
- The **auction slot** is a seat anyone can bid LP tokens for. Its holder trades at a heavily discounted fee for a period, which rewards active traders who keep the pool's price in line.
- A pool whose asset is frozen by its issuer can be frozen too.

NOSHASHI's **Pool Governance** screen shows who votes the fee, what share of LP tokens those votes represent, and who holds the auction slot, because a pool whose fee is set by one large holder behaves differently from one with a spread of voters.

KEY IDEA A pool's fee and discounts are governed by whoever holds the most LP tokens. Check who that is.

9 • Payments in depth

ABOUT 7 MINUTES

Partial payments

A Payment's **Amount** field says how much should arrive. But if the sender sets the **partial payment** flag (`tfPartialPayment`), the Amount becomes only a **maximum**. The payment can deliver far less and still finish with `tesSUCCESS`.

The authoritative figure is **delivered_amount** in the transaction's metadata. The XRPL's own documentation tells applications to credit that. Exchanges that credited Amount instead have been drained: attackers sent "large" payments that delivered almost nothing and withdrew the difference.

A REAL CASE NOSHASHI MEASURED · 28 AUGUST 2026

Amount: 999,332.87 LRC. delivered_amount: 3,958.64 LRC, about 0.4%. Result: `tesSUCCESS`. Any system crediting the stated figure would over-credit by roughly 250 times. In the same sample, 3 of 223 consecutive payments carried the partial-payment flag.

Other payment tools

TOOL	WHAT IT DOES
Destination tag	A number that tells a shared account (an exchange, a custodian) which customer the payment is for. Missing tags are a common cause of lost deposits.
Paths	Routes that let a payment convert currencies on the way, through order books and pools.

Escrow	Locks funds until a time passes or a condition is met. Escrowed funds show in an account but cannot be spent.
Checks	Like a paper cheque: the receiver chooses when to cash it.
Payment channels	Many tiny off-ledger payments settled later in one transaction.

KEY IDEA Credit what was delivered, never what was asked for.

10 · NFTs and the rights issuers keep

ABOUT 4 MINUTES

The XRPL supports native **NFTs** (non-fungible tokens, XLS-20): unique tokens for art, tickets, deeds or memberships. Each one has a 64-character **NFTokenID**, and the ID itself encodes the rules the issuer set when minting:

- **Burnable**: the issuer can destroy it, even while you own it.
- **Transferable**: if not set, you can only sell it back to the issuer.
- **Only XRP**: it can only be sold for XRP.
- **Mutable**: the issuer can change what its URI points at, so the picture or document can change after you buy it.
- **Transfer fee**: the issuer takes up to 50% of every resale.

Because these rules are inside the ID, NOSHASHI's **Token Rights** screen can decode them without even contacting the ledger.

KEY IDEA Owning an NFT does not mean the issuer is done with it. Its ID tells you what they kept.

11 • Amendments: how the rules change

ABOUT 4 MINUTES

The XRPL's rules are changed by **amendments**. A new feature is written into the server software, then validators vote. It switches on only when it keeps at least **80%** support for **two weeks** in a row. Servers that have not upgraded when an amendment activates become **amendment blocked** and stop following the ledger.

Many features in this course arrived this way: AMM (XLS-30), Clawback (XLS-39), NFTs (XLS-20), Credentials (XLS-70), Permissioned Domains (XLS-80) and Deep Freeze (XLS-77). "XLS" numbers are the public proposals behind them. NOSHASHI's Domain Grid lists which amendments are enabled.

KEY IDEA New rules need a supermajority for two weeks. What is enabled today is readable from the ledger.

12 • Compliance and regulation, made simple

ABOUT 8 MINUTES

Compliance means following the laws that apply to moving money. For anyone handling XRPL assets professionally, it comes down to two questions: **am I allowed to move this**, and **could I actually get out of it**? The first is law and policy. The second is market reality.

The words you will meet

- **KYC** (Know Your Customer): verifying who a customer is.
- **AML** (Anti-Money-Laundering): rules and monitoring to stop criminal money.
- **Sanctions screening**: checking that no party is on a government blocklist. Wallet-analytics vendors do this by studying behaviour.
- **Travel Rule** (FATF Recommendation 16): when a transfer between regulated providers is above a threshold, the sender's and receiver's details must travel with it. FATF suggests USD/EUR 1,000; the US uses USD 3,000; the EU applies it with no minimum. NOSHASHI lets you set your jurisdiction's threshold and tells you when a transfer is in scope.

- **VASP / CASP**: a virtual (or crypto) asset service provider, the regulated business the rules apply to.
- **MICA**: the EU's crypto-asset regulation, which governs stablecoin issuers such as those issuing euro e-money tokens.

What screening misses

Behaviour screening asks "is this counterparty risky?". It cannot see issuer configuration. A perfectly clean holder can be holding a token whose issuer can freeze it or claw it back. That is a property of the **asset**, not of the person, and it lives in the flags from lesson 5. NOSHASHI reads both sides.

Compliance built into the XRPL

- **Credentials (XLS-70)**: an issuer of credentials (say, a KYC provider) can attach a verifiable credential to an account. It can be checked without revealing the underlying personal data, which is called **selective disclosure**.
- **Permissioned domains (XLS-80)**: a defined area where only accounts holding the right credentials can take part.
- **Counterparty concentration**: how much of your activity goes to a few parties, measured with the **Herfindahl-Hirschman Index (HHI)**: square each party's percentage share and add them up. 10,000 means one party has everything.

KEY IDEA Know who you deal with, know what the asset allows, and keep evidence you can show later.

PART 2 · THE NOSHASHI APP

13 · What NOSHASHI is

ABOUT 4 MINUTES

NOSHASHI is a desktop app for macOS, Windows and Linux that reads the live XRP Ledger and answers the two compliance questions about any account, token, pool or payment, from the same ledger reading:

1. **Am I allowed to move this?** Issuer powers, credentials, permissioned domains, Travel Rule scope, counterparty provenance and your own policy rules.
2. **Could I actually get out of it?** Funded order-book depth, AMM pools, and how concentrated the holders are.

Other tools answer one question or the other: compliance vendors rarely read the order book, and market terminals rarely read the flags. NOSHASHI reads both from the same WebSocket in the same second, and reports them as one result.

Who it is for

WHO	WHAT THEY USE IT FOR
Compliance officers	Screening assets and settlements, Travel Rule scoping, evidence for regulators.
Trading desks and funds	Knowing which positions are exitable before size goes on.
Treasury teams	Proving who controls the treasury and what is locked, for auditors.
Token issuers	Seeing their own issuance as holders see it, and who holds it.
Auditors	Ledger-stamped, reproducible evidence of balances and control.
Anyone	Checking an address before paying it, for free.

STATUS

NOSHASHI is in **beta**, version 1.0.9. Every build is labelled beta. It provides information, not financial or legal advice.

14 · How NOSHASHI reads the ledger

ABOUT 5 MINUTES

- **Validated only.** It reads validated ledgers from mainnet, the real XRP Ledger. Nothing is estimated, simulated or backfilled.
- **Four servers, compared.** It queries four public servers by name. If they disagree, the disagreement is shown as a finding, never averaged away.
- **It says when it cannot tell.** Where a figure cannot be established (for example, too little of a supply could be read to measure concentration), the app withholds it and says so, instead of showing zero.
- **No keys, no custody.** It never asks for a secret key, never signs, never moves money. It only reads public data.
- **Stamped and fingerprinted.** Every result carries the ledger index it was read at and a **SHA-256 digest**: a fingerprint of the exact result. Change one character and the digest changes completely.
- **Local first.** Your adjudication history (up to 10,000 verdicts) is stored on your device. The AI agent runs on your device by default.
- **Self-verifying.** The app can hash its own program file, so you can confirm you are running the published build.

The calls it makes are standard XRPL server methods, listed in [Under the hood](#), so anyone can reproduce a reading with their own tools.

15 • Install and first launch

ABOUT 5 MINUTES

- 1 Open [Downloads](#) and choose your system: `.dmg` for Mac (Apple Silicon or Intel), `.msi` or `-setup.exe` for Windows, `.deb`, `.rpm` or `.AppImage` for Linux.
- 2 Install and open it. The landing screen shows the ledger live. Continue into the console.
- 3 Glance at **Mission Control** first: it confirms the app is connected and shows the latest validated ledger.

4 Use the sidebar on the left. Screens are grouped into Overview, Adjudication, Markets & Exposure, Treasury & Issuance, Record, Intelligence, Public and Growth.

5 Try **Check an Address**: paste any address starting with `r` and press `CHECK THIS ADDRESS`. The free plan includes 10 checks a month.

6 Visit **Settings**: appearance (light or dark), notifications, the global keyboard shortcut for the menu-bar HUD, and **Updates**, where new versions install in place.

7 To unlock paid screens, open **Pricing** in the app, choose a plan and sign in. **Account** holds your subscription, two-factor authentication and API keys.

Most input fields take an XRPL address (`r...`). Transaction and NFT screens take a 64-character hex hash instead. Every action button changes to "READING..." while it works and back when the result is in.

16 · Verdicts, checks and receipts

ABOUT 6 MINUTES

When NOSHASHI adjudicates something, it runs a set of **checks**, each with a severity. The overall **verdict** follows from them:

GO Every check passed. Nothing on the ledger stands in the way.

HOLD A warning-level check did not pass, or a check needs review. A person should look before moving.

NO-GO A blocking check failed. Do not move until it is resolved or formally excepted.

Each individual check reports one of five states:

PASS

Run and satisfied.

FAIL

Run and not satisfied.

REVIEW

Needs a human decision.

INSUFFICIENT_DATA

The ledger did not give enough to decide. An abstention, never a pass.

NOT_APPLICABLE

The check does not apply here.

The receipt

Every verdict produces a **cryptographic receipt**: what was checked, the ledger index, the rules version, each check's outcome and a SHA-256 digest over all of it. Copy it into your records. Anyone who later recomputes the digest over the same contents gets the same value; if anything was altered, they will not.

Your policy

Checks can be tuned by your organisation's **policy**: your thresholds and required credentials. When a verdict is shown under a policy it says so, and a verdict shown as "raw ledger facts, no policy applied" means only the built-in checks ran.

KEY IDEA Read the verdict, then open the checks: every result names the exact rule that decided it.

17 · Every screen, step by step

ABOUT 25 MINUTES · KEEP IT AS A REFERENCE

In sidebar order. Button names are shown **LIKE THIS**, exactly as they appear in the app. **FREE** marks screens that work without a paid plan.

Overview

Overview

FOR

A summary of what NOSHASHI does and what is coming.

HOW

Read it once when you start; nothing to enter.

Mission Control CONTROL ROOM

FOR

Live mainnet telemetry: latest validated ledger, network health, fee pressure and the wallet gate status.

HOW

Open it before relying on any reading. If the ledger number is climbing every few seconds, you are connected and current.

WATCH FOR

A ledger number that stops climbing means the connection dropped; readings will be stale until it resumes.

Adjudication · "am I allowed to move this?"

Verification THE GATE

FOR

Running a settlement through the compliance gate and getting GO / HOLD / NO-GO with a cryptographic receipt.

HOW

1. Enter the account to adjudicate and, optionally, the **destination account** and **amount (XRP)**.
2. Choose the target domain and required credentials if your policy uses them.
3. Run it and read the verdict, then each check.
4. Copy the receipt into your records.

YOU SEE

The verdict, every check with its state and rule, the headroom left after settlement, and the receipt.

Provenance

FOR

Where a counterparty came from: its real age and the account that first funded it.

HOW

1. Paste the counterparty's address.
2. Press **TRACE PROVENANCE**.

WATCH FOR

A days-old account funded by an unknown source deserves more questions than a years-old one funded by a known exchange.

Credentials

FOR

The XLS-70 credential registry: which credentials an account holds and who issued them, with selective disclosure.

HOW

Switch between the grid and table views; open a credential to see its issuer and status.

Domain Grid

FOR

XLS-80 permissioned domains and their policy state, plus which amendments are enabled on mainnet.

HOW

Pick a domain to see its membership rules and whether an account qualifies.

Markets & Exposure · "could I get out of it?"

Exposure Analysis RISK

FOR

One view of a position's risk: issuer freeze capability, Travel Rule scope for your jurisdiction, counterparty concentration (HHI), book exposure and a stress test.

HOW

1. Set your jurisdiction so the Travel Rule threshold is right.
2. Review the issuers, book and stress panels.

Portfolio & Radar DESK

FOR

Watching several wallets at once, with a compliance radar that flags changes and critical alerts.

HOW

1. Enter an XRPL address and a label (for example "Treasury · hot wallet").
2. Press **ADD TO BOOK**.
3. The radar keeps re-checking every wallet you add.

YOU SEE

Book balance and exposure, concentration, eligibility across domains and radar coverage.

Order Book

FOR

How much of a book's depth is backed by owners who can actually fill it (lesson 7).

HOW

1. Enter the currency code (for example `USD`) and the issuer's address.
2. Press `READ THIS BOOK`.

YOU SEE

Listed and funded depth, the unfunded share, spread and a mid price derived from the tightest uncrossed pair.

WATCH FOR

A high unfunded share means the book looks far deeper than it is.

Pool Governance AMM

FOR

Who votes an AMM pool's fee, on what share of its LP tokens, and who holds the auction slot (lesson 8).

HOW

Enter the pool's asset or account and press `READ POOL GOVERNANCE`.

Treasury & Issuance

Control Surface TREASURY

FOR

Who can move a treasury: master key status, signer list, minimum signers from summed weights, and XRP locked in reserve (lesson 6).

HOW

1. Paste the treasury address.
2. Press `READ CONTROL SURFACE`.

YOU SEE

Master key enabled or disabled, signer list members, minimum signers, spendable versus locked in reserve.

WATCH FOR

An enabled master key next to a signer list: one key can bypass the whole list.

Issuance

FOR

A token seen from the issuer's side: currencies issued, holder lines read, largest holder, concentration and enforcement history (freezes and clawbacks).

HOW

Paste the issuer's address and run it.

WATCH FOR

If too few holder lines could be read to support a concentration figure, the app withholds it rather than guess.

Authority

FOR

What an issuer can still do to its asset (lesson 5), digested so anyone can re-check it.

HOW

1. Paste the issuer address.
2. Press **CERTIFY AUTHORITY**.

YOU SEE

Six checks: freeze surrendered, not globally frozen now, open holding, no single key controls the issuer, transfer fee and supply concentration; with the verdict, ledger index and digest. The free website certificate runs the same checks.

Passport

FOR

A signed, portable record of an asset's posture to hand to someone else: verdict, outstanding supply, holders, coverage and ledger.

HOW

1. Enter the issuer address.
2. Press **GENERATE PASSPORT**.
3. Copy it as JSON or CSV, or export it.

Record

Audit Trail HISTORY

FOR

A wallet's transaction history with compliance details attached, ready to export.

HOW

Use the filter box to search by hash, account or transaction type.

Settlement

FOR

What a transaction actually delivered, not what it requested (lesson 9).

HOW

1. Paste the transaction hash (64 hex characters).
2. Press **READ SETTLEMENT**.

YOU SEE

Type, from, result, whether the partial-payment flag was set, the requested amount, the delivered amount and the fee burned.

Ledger & Policy WORKSTATION

FOR

Your adjudication history, your policy, investigations and the signed export.

HOW

1. Browse past verdicts with **PREV** and **NEXT**.
2. Open **Policy** to draft rule changes; drafts can be simulated before they go live.
3. Open **Cases** for investigations.
4. Press **EXPORT** for a file with a SHA-256 chain-of-custody signature; the screen explains how a recipient verifies it.

Intelligence

Agent

FOR

An AI analyst and support desk. See lesson 19.

Ledger Garden

FOR

Exploring visually: walk from issuer to asset to holder to transaction to evidence, then ask the agent about what you found.

HOW

Enter an `I...` address or a 64-character hash and follow the branches.

Public tools

Check an Address

FREE · 10 A MONTH

FOR

Anyone: read what the ledger publishes about an account before you pay it.

HOW

1. Paste the address.

2. Press `CHECK THIS ADDRESS`.

REMEMBER

"Nothing recorded against it" is not the same as "safe". The screen reports facts and never gives advice.

Inbox CLAIMS

FOR

Tokens and claims strangers have sent to an account, and whether they are real.

HOW

Paste the account address and press `READ CLAIMS`.

WATCH FOR

A familiar ticker from an issuer with zero obligations is an impersonation. It is flagged as uncashable, not shown as a balance.

Token Rights NFT

FOR

What an NFT's issuer can still do after you own it (lesson 10).

HOW

Paste the 64-character NFTokenID and press **READ TOKEN RIGHTS**.

Ledger Sync FREE

FOR

What four public servers each report right now, and where they disagree. No account needed.

HOW

Open it; press **QUERY AGAIN** to refresh.

Learn

FOR

Short animated explainers inside the app: the gate, freeze rights, book depth and the address check.

Growth, account and extras

Growth

FOR

Drafts for posts and pitches built only from figures the app measured, each with its date. There is no publish button: you post them yourself.

Pricing and Account

FOR

Plans, checkout and verification credits; your subscription, two-factor authentication and API keys.

Menu-bar HUD macOS

FOR

A small live panel in the Mac menu bar with a ticker, so you can glance at the ledger without opening the full app.

HOW

Open it with the global keyboard shortcut set in Settings, or click the tray icon.

18 · For organisations and developers

ABOUT 5 MINUTES

Organisation policy with four-eyes approval

An organisation can share one policy. A change is saved as a **draft**, can be **simulated** against past verdicts to see what it would have changed, and goes **pending** when submitted. A second person must approve it before it is active (**four-eyes**). Every step is logged.

Exceptions and evidence requests

When a NO-GO must be overridden for a good reason, a user requests an **exception**. A reviewer can approve, reject, or **request more evidence** first. The decision and its reasons are kept with the verdict.

Investigations

An **investigation** (a case) collects verdicts, readings and notes about one issuer or counterparty. Its log is **hash-chained**: each entry includes the fingerprint of the one before, so entries cannot be quietly edited or removed. Cases can be shared across an organisation.

Webhooks and the Compliance API

Webhooks send signed notifications of events (for example, a policy activated) to your own systems. The **Compliance API** offers read access to verdicts and readings for

integration; see [the API docs](#).

19 • The Agent and the Observer

ABOUT 4 MINUTES

The **Agent** is an AI assistant built into the app. By default it runs on your own device through a local model (Ollama or LM Studio), so nothing leaves your machine. You can connect a remote provider with your own API key instead.

- **Compliance mode**: ask about a reading, a verdict or a rule, and it answers from what the app actually measured.
- **Support mode**: help with using the app, with human escalation when needed.
- **Observer mode**: watches the issuers and wallets you care about and raises alerts when something changes, such as an issuer flag flipping. One click opens an investigation with the evidence attached.
- **Governance**: rules for what the agent may and may not do.

USE IT WISELY

The agent explains; the ledger decides. Always confirm important points against the readings and receipts themselves.

PART 3 · WORKFLOWS: REAL TASKS, START TO FINISH

Eight things you can do today

EACH TAKES 2 TO 10 MINUTES

A • Before paying someone new

- 1 Open **Check an Address**, paste their address, press **CHECK THIS ADDRESS**.

2 Open **Provenance**, press **TRACE PROVENANCE**: how old is the account and who funded it?

3 If they are an exchange, confirm the destination tag you were given.

4 Decide knowingly; keep the readings if the amount matters.

B · Before accepting or buying a token

1 Find the token's issuer address, not just its name.

2 **Authority** → **CERTIFY AUTHORITY**: can the issuer freeze, claw back or act with one key? What fee does it charge?

3 **Order Book** → enter the code and issuer → **READ THIS BOOK**: how much funded depth is there to exit?

4 **Issuance**: how concentrated are the holders?

5 **Passport** → **GENERATE PASSPORT** to keep or share the record.

C · Checking that a payment really arrived

1 Copy the transaction hash from the sender or your history.

2 **Settlement** → paste it → **READ SETTLEMENT**.

3 Compare requested with delivered. If the partial-payment flag is set, credit only the delivered amount.

D · Proving who controls a treasury (for an auditor)

1 **Control Surface** → treasury address → **READ CONTROL SURFACE**.

2 Record the minimum signers, whether the master key is disabled, and what is locked in reserve.

3 **Ledger & Policy** → **EXPORT** for a signed file. Send it with the ledger index; the auditor can re-check both.

E · Running a settlement through your policy

1 **Verification** → enter the account, destination and amount.

2 Run it. Read GO / HOLD / NO-GO, then open each check.

3 On NO-GO with a legitimate reason, request an exception; a reviewer decides, possibly after asking for evidence.

4 Copy the receipt into your records.

F · Watching issuers you depend on

1 Add the wallets and issuers to **Portfolio & Radar** with **ADD TO BOOK**.

2 Turn on the **Observer** in the Agent.

3 When a flag changes, open the alert and start an investigation in one click.

G · Sorting out unexpected tokens in your wallet

1 **Inbox** → your address → **READ CLAIMS**.

2 Anything flagged uncashable is an impersonation. Do not interact with it or visit links in its memo.

H · Changing your organisation's rules safely

1 **Ledger & Policy** → **Policy** → create a draft with the new threshold.

2 Simulate it against past verdicts to see what would have changed.

3 Submit it; a second person approves it (four-eyes). It is logged either way.

Free tools on this website

- [Authority certificate](#): paste any issuer address and get the same six checks as the app's Authority screen, with the ledger index and digest. Free and no account needed.
- [Misread cases](#): real examples where a quick reading of the ledger gives the wrong answer.
- [Research](#): measurements and methods, each with its date and ledger.
- [Docs](#): getting started, architecture, receipts, policies, webhooks, the API and troubleshooting.
- [Plain-language guide](#): a shorter introduction.
- [Status](#) and [Progress](#): service health and release notes.

Plans, trust and what's next

PLAN	FOR	PRICE
Free	One address, Check an Address (10 a month), Ledger Sync, the website tools	\$0
Pro	A desk: every screen, per seat	\$749 / seat / month
Institutional	Regulated teams, unlimited seats, organisation features	\$4,000 / month

Enterprise ·
Strategic

Larger organisations and infrastructure
partners

from \$120,000 / year

Current details are on [Pricing](#). The code is public on GitHub, every reading is ledger-stamped and fingerprinted, and security details are on [Trust & security](#).

Coming next

Opt-in crash reporting, a phone-sized layout, translations into seven languages, and an iPhone and iPad version (it already compiles; see the [iOS notes](#)).

Questions people ask

▼ Do I need to own crypto or connect a wallet?

No. NOSHASHI only reads public ledger data. It never asks for a secret key and cannot move funds.

▼ Does a GO mean something is safe?

No. GO means nothing on the ledger stands in the way under the checks that ran. It says nothing about the business behind an issuer, prices or the future.

▼ Why is a well-known issuer NO-GO?

Usually because one key can sign for it, or it keeps freeze rights. Those are normal for many regulated issuers. The verdict states the fact so you decide knowingly; it is not an accusation.

▼ Why does a figure say "insufficient data" instead of a number?

Because the ledger did not return enough to support one. NOSHASHI will not guess and will not show zero for "unknown".

▼ Can I check a reading myself?

Yes. Ask any XRPL server for the same data at the same ledger index, or re-run the free certificate. Recomputing the digest over the same contents gives the same value.

▼ Where is my data stored?

Your verdict history and the local agent stay on your device. Account, plan and organisation features use the NOSHASHI service. The website certificate stores nothing.

▼ Is this financial or legal advice?

No. It reports ledger facts and what they imply. Decisions remain yours and your advisers’.

PART 4 · KNOWLEDGE CHECK

Knowledge check

55 QUESTIONS IN 5 ROUNDS · YOUR ANSWERS STAY IN YOUR BROWSER

Rounds 1 to 3 cover the ledger, markets and payments, and the app. Round 4 tests the vocabulary. Round 5 asks how NOSHASHI helps institutions and enterprises: treasuries, compliance teams, desks, payments firms, issuers and auditors.

ROUND 1 · THE XRP LEDGER

1. What makes a ledger "validated"?

- ☐ It is older than a day
- ☐ At least 80% of trusted validators agreed on it, so it is final
- ☐ NOSHASHI approved it
- ☐ It contains only XRP payments

▼ WORDS USED HERE (2)

Validator	A server that votes on which ledger is the true next one.
Ledger	One numbered snapshot of the whole XRPL, produced every 3 to 5 seconds.

Answer: At least 80% of trusted validators agreed on it, so it is final. Validators vote; when 80% of a server's trusted list agree, the ledger is validated and can never change.

2. Roughly how often does the XRPL produce a new ledger?

- ☐ Every 10 minutes
- ☐ Every 3 to 5 seconds
- ☐ Once a day
- ☐ Only when someone pays

▼ WORDS USED HERE (3)

Ledger index	A ledger's number. Fixes a reading to a moment anyone can re-check.
Ledger	One numbered snapshot of the whole XRPL, produced every 3 to 5 seconds.
XRPL	The XRP Ledger.

Answer: Every 3 to 5 seconds. A new ledger closes every 3 to 5 seconds, each with a higher ledger index.

3. What happens to the XRP paid as a transaction cost?

- ☐ It goes to validators
- ☐ It is destroyed
- ☐ It goes to Ripple
- ☐ It returns to the sender

▼ WORDS USED HERE (2)

Transaction cost	The small XRP fee each transaction pays, which is destroyed.
XRP	The XRP Ledger's own currency.

Answer: It is destroyed. The cost is burned, which protects the network from spam without paying anyone.

4. Why can't all of an account's XRP always be spent?

- ☐ Because of the reserve
- ☐ Because XRP expires
- ☐ Because of the transfer fee
- ☐ Because of consensus

▼ WORDS USED HERE (4)

Owner reserve	XRP locked for each object an account owns, such as a trust line or offer.
Account	A place on the ledger that holds XRP and tokens, identified by an address starting with r.
Reserve	XRP an account must keep locked: a base reserve plus an owner reserve per object.
XRP	The XRP Ledger's own currency.

Answer: Because of the reserve. Every account locks a base reserve plus an owner reserve per object it owns.

5. An account's Sequence number is 91,000,000. What does that tell you?

- ☐ It made 91 million transactions
- ☐ Its counter started near the ledger index when it was created
- ☐ It is blackholed
- ☐ Nothing at all

▼ WORDS USED HERE (3)

Ledger index	A ledger's number. Fixes a reading to a moment anyone can re-check.
Sequence	An account's transaction counter, starting at its creation ledger index.
Account	A place on the ledger that holds XRP and tokens, identified by an address starting with r.

Answer: Its counter started near the ledger index when it was created. Sequence starts at the creation ledger index, not zero, so it hints at age, not activity.

6. A transaction result starts with tec. What does that mean?

- ☐ Success
- ☐ It failed but was recorded and the fee was charged
- ☐ It was never submitted
- ☐ It is still pending

▼ WORDS USED HERE (3)

Transaction	A signed instruction that changes the ledger.
Ledger	One numbered snapshot of the whole XRPL, produced every 3 to 5 seconds.
tec	A result prefix: failed, but recorded and the fee charged.

Answer: It failed but was recorded and the fee was charged. tec means claimed cost only: the transaction failed but is in the ledger and the fee is gone.

7. Two tokens are both called USD. What tells them apart?

- ☐ Their price
- ☐ The issuer address
- ☐ The colour in your wallet
- ☐ Nothing, they are the same

▼ WORDS USED HERE (3)

Address	An account's public identifier, 25 to 35 characters starting with r.
Issuer	The account that creates a token and sets its rules.
Token	An asset on the XRPL other than XRP, created by an issuer.

Answer: The issuer address. Anyone can use the code USD. The issuer address is the token's identity.

8. What must you open before you can hold an issued token?

- ☐ An escrow
- ☐ A trust line to the issuer
- ☐ An AMM pool
- ☐ A signer list

▼ WORDS USED HERE (4)

Trust line	The link you open to an issuer to hold its token.
Issuer	The account that creates a token and sets its rules.
Limit	The most of a token you are willing to hold on a trust line.
Token	An asset on the XRPL other than XRP, created by an issuer.

Answer: A trust line to the issuer. A trust line records your limit and balance with that issuer.

9. What does an issuer's gateway_balances show?

- ☐ Its XRP balance
- ☐ Its obligations: how much of each token others hold
- ☐ Its fees
- ☐ Its validators

▼ WORDS USED HERE (4)

gateway_balances	The server call that returns an issuer's obligations.
Obligations	The total of an issuer's tokens held by others: its real circulating supply.
Issuer	The account that creates a token and sets its rules.
Token	An asset on the XRPL other than XRP, created by an issuer.

Answer: Its obligations: how much of each token others hold. It is the real circulating supply of each token the issuer has issued.

10. Which amendment rule is correct?

- ☐ 51% for a day
- ☐ 80% of validators for two weeks
- ☐ Any validator can switch it on
- ☐ Ripple decides alone

▼ WORDS USED HERE (2)

Amendment	A change to the XRPL's rules, enabled after 80% of validators support it for two weeks.
Validator	A server that votes on which ledger is the true next one.

Answer: 80% of validators for two weeks. An amendment needs at least 80% support held for two weeks.

ROUND 2 · ISSUERS, MARKETS AND PAYMENTS

11. An issuer has set No Freeze. What does that mean?

- ☐ It can freeze whenever it likes
- ☐ It has permanently given up freezing
- ☐ The token is frozen now
- ☐ Only XRP can be frozen

▼ WORDS USED HERE (2)

No Freeze	A permanent promise by an issuer never to freeze.
Issuer	The account that creates a token and sets its rules.

Answer: It has permanently given up freezing. No Freeze is permanent and cannot be switched back off.

12. Which power lets an issuer take tokens back from your account?

- ☐ Transfer fee
- ☐ Require Auth
- ☐ Clawback
- ☐ Destination tag

▼ WORDS USED HERE (5)

Trust line	The link you open to an issuer to hold its token.
Clawback	An issuer power to take tokens back from a holder. Must be enabled before any trust lines exist (XLS-39).
Account	A place on the ledger that holds XRP and tokens, identified by an address starting with r.
Issuer	The account that creates a token and sets its rules.
Token	An asset on the XRPL other than XRP, created by an issuer.

Answer: Clawback. Clawback, which can only be enabled before the issuer has any trust lines.

13. What does a deep freeze add to a normal freeze?

- ☐ Nothing
- ☐ The holder can neither send nor receive the token
- ☐ It freezes XRP too
- ☐ It is temporary

▼ WORDS USED HERE (3)

Deep freeze	A freeze where the holder can neither send nor receive the token (XLS-77).
Freeze	An issuer power to stop a holder sending its token on.
Token	An asset on the XRPL other than XRP, created by an issuer.

Answer: The holder can neither send nor receive the token. Deep freeze (XLS-77) blocks receiving as well as sending.

14. A signer list has weights 3, 2, 2, 1, 1 and a quorum of 5. What is the fewest number of signers needed?

- ☐ Five
- ☐ Three
- ☐ Two
- ☐ One

▼ WORDS USED HERE (2)

Signer list	The keys and weights used for multi-signature.
Quorum	The minimum total signing weight needed to act for an account.

Answer: Two. The weight-3 and a weight-2 signer reach 5 together. Count weights, not people.

15. What makes an issuer blackholed?

- ☐ It was hacked
- ☐ Master key disabled, regular key set to an address nobody controls, no signer list
- ☐ It has no holders
- ☐ It charges no fee

▼ WORDS USED HERE (5)

Regular key	A second key an account nominates to sign for it.
Signer list	The keys and weights used for multi-signature.
Blackholed	An account nobody can sign for again: master key disabled, regular key unusable, no signer list.
Master key	An account's original key pair. It can be disabled.
Address	An account's public identifier, 25 to 35 characters starting with r.

Answer: Master key disabled, regular key set to an address nobody controls, no signer list. Nobody can ever sign for it again.

16. What is an unfunded offer?

- ☐ An offer with no price
- ☐ An offer still listed although its owner cannot fill it
- ☐ An offer in XRP
- ☐ A cancelled offer

▼ WORDS USED HERE (2)

Unfunded offer	An offer still listed although its owner cannot fill it.
Offer	An order posted on the DEX to trade one asset for another.

Answer: An offer still listed although its owner cannot fill it. Offers stay listed after owners stop holding the funds.

17. Three offers list 10,000 each; owners hold 10,000, 500 and 0. What is the funded depth?

- ☐ 30,000
- ☐ 10,500
- ☐ 10,000
- ☐ 500

▼ WORDS USED HERE (3)

Funded depth	Depth backed by owners who can actually deliver.
Listed depth	All the depth shown on a book, including offers nobody can fill.
Offer	An order posted on the DEX to trade one asset for another.

Answer: 10,500. $10,000 + 500 + 0 = 10,500$. Listed depth would say 30,000.

18. Which field does book_offers add when an owner cannot cover an offer?

- ☐ delivered_amount
- ☐ taker_gets_funded
- ☐ TransferRate
- ☐ Sequence

▼ WORDS USED HERE (3)

taker_gets_funded	The field book_offers adds when an owner cannot cover an offer.
book_offers	The server call that returns an order book's offers.
Offer	An order posted on the DEX to trade one asset for another.

Answer: taker_gets_funded. taker_gets_funded shows what the owner can actually deliver.

19. Who votes on an AMM pool's trading fee?

- ☐ Validators
- ☐ LP token holders, weighted by holdings
- ☐ The issuer
- ☐ Nobody, it is fixed

▼ WORDS USED HERE (2)

LP token	A token received for adding assets to an AMM pool; carries a fee vote.
AMM	Automated market maker: a pool of two assets that trades automatically (XLS-30).

Answer: LP token holders, weighted by holdings. Votes are weighted by LP tokens, so large holders carry more weight.

20. A payment shows `tesSUCCESS` with the partial-payment flag. What should you credit?

- ☐ The Amount field
- ☐ `delivered_amount`
- ☐ The fee
- ☐ Nothing

▼ WORDS USED HERE (3)

<code>delivered_amount</code>	The amount a payment actually delivered, in its metadata. The figure to credit.
<code>tesSUCCESS</code>	The success result. It does not mean the full Amount arrived.
Flag	An on/off setting on an account or trust line.

Answer: `delivered_amount`. With the flag, Amount is only a maximum. `delivered_amount` is what arrived.

ROUND 3 · COMPLIANCE AND THE APP

21. What does the Travel Rule require?

- ☐ Slow payments
- ☐ Sender and receiver details travel with transfers above a threshold
- ☐ Freezing all tokens
- ☐ Only domestic payments

▼ WORDS USED HERE (1)

Travel Rule

FATF Recommendation 16: sender and receiver details travel with transfers above a threshold.

Answer: Sender and receiver details travel with transfers above a threshold. FATF Recommendation 16. Thresholds vary: USD/EUR 1,000 suggested, USD 3,000 in the US, none in the EU.

22. Why can't behaviour screening alone catch issuer risk?

- ☐ It is too slow
- ☐ Issuer powers are a property of the asset, not of the person holding it
- ☐ Issuers are always safe
- ☐ Screening reads flags already

▼ WORDS USED HERE (3)

Issuer

The account that creates a token and sets its rules.

Token

An asset on the XRPL other than XRP, created by an issuer.

Flag

An on/off setting on an account or trust line.

Answer: Issuer powers are a property of the asset, not of the person holding it. A clean holder can hold a clawback-enabled token. Only the issuer's flags reveal that.

23. An HHI of 10,000 means what?

- ☐ Perfectly spread out
- ☐ One party has everything
- ☐ Ten thousand holders
- ☐ The fee is 100%

▼ WORDS USED HERE (2)

Concentration	How much of an asset or activity sits with a few parties; measured with the HHI.
HHI	Herfindahl-Hirschman Index: the sum of squared percentage shares; 10,000 is total concentration.

Answer: One party has everything. HHI sums squared percentage shares; $100^2 = 10,000$ is total concentration.

24. What are the two questions NOSHASHI answers?

- ☐ Price and volume
- ☐ Am I allowed to move this, and could I actually get out of it?
- ☐ Who owns XRP, and when to buy
- ☐ Which wallet is fastest

▼ WORDS USED HERE (2)

Compliance	Following the laws and rules that apply to moving money.
Ledger	One numbered snapshot of the whole XRPL, produced every 3 to 5 seconds.

Answer: Am I allowed to move this, and could I actually get out of it?. Compliance and exit liquidity, from the same ledger reading.

25. A check shows `INSUFFICIENT_DATA`. How should you treat it?

- ☐ As a pass
- ☐ As a fail
- ☐ As an abstention: the ledger did not give enough to decide
- ☐ As a bug

▼ WORDS USED HERE (2)

INSUFFICIENT_DATA

A check state meaning the ledger did not give enough to decide. Never a pass.

Ledger

One numbered snapshot of the whole XRPL, produced every 3 to 5 seconds.

Answer: As an abstention: the ledger did not give enough to decide. It is never a pass. NOSHASHI would rather say it cannot tell than guess.

26. An issuer fails "no single key controls the issuer". Which verdict follows?

- ☐ GO
- ☐ HOLD
- ☐ NO-GO
- ☐ NOT_APPLICABLE

▼ WORDS USED HERE (2)

Issuer

The account that creates a token and sets its rules.

NO-GO

A verdict meaning a blocking check failed.

Answer: NO-GO. That check is blocking, so a failure gives NO-GO.

27. Which screen tells you who can really move a treasury?

- ☐ Order Book
- ☐ Control Surface
- ☐ Inbox
- ☐ Ledger Sync

▼ WORDS USED HERE (1)

Flag

An on/off setting on an account or trust line.

Answer: Control Surface. Control Surface derives minimum signers from weights and flags a master-key bypass.

28. You need to check a payment really arrived. Which screen and button?

- ☐ Settlement · READ SETTLEMENT
- ☐ Passport · GENERATE PASSPORT
- ☐ Authority · CERTIFY AUTHORITY
- ☐ Growth

Answer: Settlement · READ SETTLEMENT. Settlement compares requested with delivered.

29. Why does every reading carry a ledger index and a SHA-256 digest?

- ☐ Decoration
- ☐ So anyone can re-check the same moment and prove the record was not changed
- ☐ To make files bigger
- ☐ It is required by the Travel Rule

▼ WORDS USED HERE (2)

Digest (SHA-256)	A fixed-length fingerprint of data. Change anything and the digest changes completely.
Ledger index	A ledger's number. Fixes a reading to a moment anyone can re-check.

Answer: So anyone can re-check the same moment and prove the record was not changed. The index fixes the moment; the digest proves the contents are unchanged.

30. What does four-eyes approval mean for a policy change?

- ☐ It needs four approvers
- ☐ A second person must approve before it goes live
- ☐ It is reviewed by four validators
- ☐ It lasts four days

▼ WORDS USED HERE (2)

Four-eyes	A control where a second person must approve a change.
Policy	An organisation's own thresholds and rules applied to verdicts.

Answer: A second person must approve before it goes live. One person drafts and submits; a different person approves.

Each question is a word from the word list. Pick the meaning that fits.

31. What is a drop?

- ☐ A token removed by clawback
- ☐ The smallest unit of XRP: 1 XRP = 1,000,000 drops
- ☐ A ledger that failed consensus
- ☐ The fee an issuer charges

Answer: The smallest unit of XRP: 1 XRP = 1,000,000 drops. Balances and fees are recorded in drops. A reserve of 1 XRP is 1,000,000 drops.

32. What is the UNL (Unique Node List)?

- ☐ The validators a server trusts not to collude
- ☐ A list of banned addresses
- ☐ Every unfunded offer on the DEX
- ☐ The tokens NOSHASHI has approved

Answer: The validators a server trusts not to collude. A ledger is validated when 80% of the validators on a server's UNL agree.

33. What are an issuer's obligations?

- ☐ Its legal duties under MiCA
- ☐ The XRP it keeps in reserve
- ☐ The total of its tokens held by others: its real circulating supply
- ☐ Offers it is forced to fill

Answer: The total of its tokens held by others: its real circulating supply. gateway_balances reports them per currency. It is the figure concentration is measured against.

34. What is a regular key?

- ☐ The first key an account ever had
- ☐ A validator's signing key
- ☐ The key that opens a permissioned domain
- ☐ A second key an account nominates to sign for it

Answer: A second key an account nominates to sign for it. If a regular key is set, it can sign even when the master key is disabled. That is why blackholing needs both handled.

35. What does Require Auth mean?

- ☐ You need the issuer's approval to hold its token
- ☐ Every payment needs two signatures
- ☐ The token can only be sold for XRP
- ☐ The issuer must approve every trade

Answer: You need the issuer's approval to hold its token. It is an issuer setting. It is one of the six authority checks, a warning when it is on.

36. What is a transaction's metadata?

- ☐ The memo the sender typed
- ☐ Its price in US dollars
- ☐ The record of what it actually did, attached once it is validated
- ☐ The list of validators that voted

Answer: The record of what it actually did, attached once it is validated. Metadata is where delivered_amount lives, which is why NOSHASHI reads it rather than the Amount field.

37. In NOSHASHI, what is provenance?

- ☐ Where an account came from: its age and first funder
- ☐ The country an issuer is based in
- ☐ The price history of a token
- ☐ Who wrote the rules version

Answer: Where an account came from: its age and first funder. The Provenance screen traces it with TRACE PROVENANCE, useful before paying a new counterparty.

38. What is a destination tag?

- ☐ A label for NFTs
- ☐ A number telling a shared account which customer a payment is for
- ☐ A tag that freezes a payment
- ☐ The name of a trust line

Answer: A number telling a shared account which customer a payment is for. Exchanges hold many customers in one address. Without the right tag, a payment arrives but nobody knows whose it is.

39. What is a credential (XLS-76)?

- ☐ A password for the app
- ☐ A paid NOSHASHI plan
- ☐ A validator's licence
- ☐ An on-ledger proof that an account passed a check

Answer: An on-ledger proof that an account passed a check. An issuer of credentials (for example a KYC provider) records the proof on the ledger. Permissioned domains (XLS-80) can require it.

40. What does selective disclosure mean?

- ☐ Publishing every customer's data
- ☐ Proving a credential exists without revealing the personal data behind it
- ☐ Only disclosing profits
- ☐ Hiding transactions from validators

Answer: Proving a credential exists without revealing the personal data behind it. It lets a counterparty confirm a check was passed without receiving the documents.

41. What is the spread?

- ☐ The gap between the best bid and the best ask
- ☐ How many holders a token has
- ☐ The AMM trading fee
- ☐ The time between ledgers

Answer: The gap between the best bid and the best ask. A wide spread means trading costs more. It says nothing about how much can be filled; funded depth does.

42. In NOSHASHI, what is an exception?

- ☐ A crash report
- ☐ A check that was skipped
- ☐ An approved, recorded override of a NO-GO for a stated reason
- ☐ A payment that bounced

Answer: An approved, recorded override of a NO-GO for a stated reason. It never erases the NO-GO. The override, its reason and its reviewer are kept alongside it.

ROUND 5 · HOW IT HELPS INSTITUTIONS AND ENTERPRISES

43. A listed company holds XRP in its treasury. Its auditor asks whether the year-end value is defensible. How does NOSHASHI help?

- ☐ It sets the price itself
- ☐ It measures how much could really be sold in the market at a stated ledger index, so the auditor can re-check the figure
- ☐ It guarantees the auditor signs off
- ☐ It moves the XRP into cold storage

▼ WORDS USED HERE (3)

Funded depth	Depth backed by owners who can actually deliver.
Ledger index	A ledger's number. Fixes a reading to a moment anyone can re-check.
XRP	The XRP Ledger's own currency.

Answer: It measures how much could really be sold in the market at a stated ledger index, so the auditor can re-check the figure. Auditors need evidence of the principal market. Funded depth stamped with a ledger index is evidence anyone can reproduce.

44. A compliance officer is asked to approve holding a new stablecoin. Which reading answers 'could the issuer freeze or claw back what we hold?'

- ☐ Growth
- ☐ Ledger Sync
- ☐ Authority: CERTIFY AUTHORITY reads the issuer's flags and returns GO, HOLD or NO-GO
- ☐ The HUD

▼ WORDS USED HERE (5)

Concentration	How much of an asset or activity sits with a few parties; measured with the HHI.
Require Auth	An issuer setting: you need its approval to hold its token.
Transfer fee	A percentage (0% to 100%) an issuer charges when holders send its token to each other.
Compliance	Following the laws and rules that apply to moving money.
Stablecoin	A token meant to hold a steady value, such as one US dollar.

Answer: Authority: CERTIFY AUTHORITY reads the issuer's flags and returns GO, HOLD or NO-GO. Freeze, clawback, Require Auth, key control, transfer fee and concentration are properties of the asset. Screening the people involved cannot see them.

45. An examiner asks a bank to show why a transfer was allowed eight months ago.

What does NOSHASHI give it?

- ☐ A screenshot
- ☐ The receipt: ledger index, rules version, checks and digest, so the decision can be reproduced exactly
- ☐ A phone number for support
- ☐ Nothing; the ledger has changed since

▼ WORDS USED HERE (3)

Digest (SHA-256)	A fixed-length fingerprint of data. Change anything and the digest changes completely.
Ledger index	A ledger's number. Fixes a reading to a moment anyone can re-check.
Receipt	NOSHASHI's record of a verdict: ledger index, rules version, checks and digest.

Answer: The receipt: ledger index, rules version, checks and digest, so the decision can be reproduced exactly. The ledger index fixes the moment. The digest proves the record was not edited afterwards.

46. Why does four-eyes approval matter to an enterprise?

- ☐ It makes changes faster
- ☐ No single employee can quietly change the rules that decide what the firm may do
- ☐ It is needed to install the app
- ☐ It lowers the price

▼ WORDS USED HERE (1)

Four-eyes

A control where a second person must approve a change.

Answer: No single employee can quietly change the rules that decide what the firm may do. Segregation of duties is a basic control auditors and regulators expect. Drafts can be simulated before anyone approves them.

47. A trading desk is about to take a large position in an XRPL token. What risk does

NOSHASHI check that a price chart hides?

- ☐ Whether the logo is official
- ☐ Whether the listed depth includes unfunded offers, so the way out is far thinner than it looks
- ☐ The token's age in days
- ☐ The colour of the chart

▼ WORDS USED HERE (5)

Unfunded offer	An offer still listed although its owner cannot fill it.
Funded depth	Depth backed by owners who can actually deliver.
Listed depth	All the depth shown on a book, including offers nobody can fill.
Token	An asset on the XRPL other than XRP, created by an issuer.
XRPL	The XRP Ledger.

Answer: Whether the listed depth includes unfunded offers, so the way out is far thinner than it looks. In the 8 September sample, listed depth was a median 4.4 times funded depth within the band.

48. A payments company receives XRPL payments on behalf of customers. Which two details stop it crediting the wrong amount or the wrong customer?

- ☐ delivered_amount and the destination tag
- ☐ Sequence and the fee
- ☐ The currency code and the spread
- ☐ The UNL and the ledger index

▼ WORDS USED HERE (4)

delivered_amount	The amount a payment actually delivered, in its metadata. The figure to credit.
Destination tag	A number telling a shared account which customer a payment is for.
Partial payment	A payment whose Amount is only a maximum; it can deliver less and still succeed.
XRPL	The XRP Ledger.

Answer: delivered_amount and the destination tag. Credit delivered_amount, never the Amount field of a partial payment, and match the destination tag to the customer.

49. An enterprise already runs its own case management and alerting. How does NOSHASHI fit in?

- ☐ It replaces them
- ☐ It does not; it is desktop only
- ☐ Signed webhooks push events to your systems, and the Compliance API gives read access to verdicts and readings
- ☐ It emails a spreadsheet each week

▼ WORDS USED HERE (2)

Compliance API

NOSHASHI's read-only interface for verdicts and readings.

Webhook

A signed notification NOSHASHI sends to your own system when an event happens.

Answer: Signed webhooks push events to your systems, and the Compliance API gives read access to verdicts and readings. Webhooks are signed so your systems can prove they came from NOSHASHI.

50. Why is a hash-chained investigation log valuable to a regulated firm?

- ☐ It saves disk space
- ☐ Entries cannot be quietly edited or removed, so the record of what the team knew, and when, holds up under review
- ☐ It hides the log from auditors
- ☐ It is faster to search

▼ WORDS USED HERE (1)

Investigation

A case in NOSHASHI collecting verdicts, readings and notes, with a hash-chained log.

Answer: Entries cannot be quietly edited or removed, so the record of what the team knew, and when, holds up under review. Each entry carries the fingerprint of the one before. Changing any entry breaks every link after it.

51. A NO-GO blocks a trade the business has a legitimate reason to make. What is the controlled way forward?

- ☐ Turn NOSHASHI off
- ☐ Edit the receipt
- ☐ Request an exception; a reviewer approves, rejects or asks for more evidence, and the decision stays with the verdict
- ☐ Ask the issuer to change its flags

▼ WORDS USED HERE (2)

Exception

An approved, recorded override of a NO-GO for a stated reason.

NO-GO

A verdict meaning a blocking check failed.

Answer: Request an exception; a reviewer approves, rejects or asks for more evidence, and the decision stays with the verdict. Overrides happen in the open, with a reason and a second person, which is what an examiner looks for.

52. A token issuer wants to show institutional buyers exactly what they would be exposed to. What can it share?

- ☐ Its marketing deck
- ☐ A link to the free authority certificate for its issuer address, which anyone can re-check
- ☐ Its private keys
- ☐ Its bank statements

▼ WORDS USED HERE (5)

Digest (SHA-256)	A fixed-length fingerprint of data. Change anything and the digest changes completely.
Ledger index	A ledger's number. Fixes a reading to a moment anyone can re-check.
Account	A place on the ledger that holds XRP and tokens, identified by an address starting with r.
Address	An account's public identifier, 25 to 35 characters starting with r.
Issuer	The account that creates a token and sets its rules.

Answer: A link to the free authority certificate for its issuer address, which anyone can re-check. The certificate page at </certificate/> runs the same six checks with the ledger index and digest, free and without an account.

53. Which plan is built for regulated teams that need organisation features and unlimited seats?

- ☐ Free
- ☐ Pro
- ☐ Institutional
- ☐ None; each seat is separate

Answer: Institutional. Institutional is \$4,000 a month. Enterprise and Strategic, from \$120,000 a year, are for larger organisations and infrastructure partners.

54. What does answering both questions from one ledger reading save a compliance team?

- ☐ Nothing
- ☐ Reconciling a screening tool and a market terminal that looked at the ledger at different moments
- ☐ Hiring an auditor
- ☐ Paying transaction costs

▼ WORDS USED HERE (2)

Compliance

Following the laws and rules that apply to moving money.

Ledger

One numbered snapshot of the whole XRPL, produced every 3 to 5 seconds.

Answer: Reconciling a screening tool and a market terminal that looked at the ledger at different moments. When compliance and liquidity come from the same WebSocket in the same second, the file tells one consistent story.

55. NOSHASHI is in beta. What should an institution take from that today?

- ☐ The readings are invented
- ☐ It provides information, not financial or legal advice; re-measure a reading before relying on it for a decision
- ☐ It cannot be installed
- ☐ Beta means free forever

▼ WORDS USED HERE (1)

Ledger

One numbered snapshot of the whole XRPL, produced every 3 to 5 seconds.

Answer: It provides information, not financial or legal advice; re-measure a reading before relying on it for a decision. Every build is labelled beta. Every reading is real and stamped with its ledger, so re-checking it is straightforward.

PART 5 · WORD LIST

Word list

EVERY TERM IN THIS COURSE, A TO Z

Account

A place on the ledger that holds XRP and tokens, identified by an address starting with r.

AccountSet

The transaction that changes an account's settings and flags.

Address

An account's public identifier, 25 to 35 characters starting with r.

Adjudication

NOSHASHI's process of checking something against rules and returning a verdict.

Amendment

A change to the XRPL's rules, enabled after 80% of validators support it for two weeks.

Amendment blocked

A server that has not upgraded when an amendment activates, and stops following the ledger.

AML

Anti-Money-Laundering: rules to stop criminal money moving.

AMM

Automated market maker: a pool of two assets that trades automatically (XLS-30).

Auction slot

A seat in an AMM pool that gives its holder discounted trading for a time.

Auto-bridging

Routing a token-to-token trade through XRP when that gives a better price.

Blackholed

An account nobody can sign for again: master key disabled, regular key unusable, no signer list.

Blocking check

A check whose failure gives NO-GO.

book_offers

The server call that returns an order book's offers.

CASP / VASP

A crypto (or virtual) asset service provider: a regulated business handling digital assets.

Check (payment)

A deferred payment the receiver cashes when they choose.

Clawback

An issuer power to take tokens back from a holder. Must be enabled before any trust lines exist (XLS-39).

Closed ledger

A ledger that has stopped taking transactions but is not yet validated.

Compliance

Following the laws and rules that apply to moving money.

Compliance API

NOSHASHI's read-only interface for verdicts and readings.

Concentration

How much of an asset or activity sits with a few parties; measured with the HHI.

Consensus

How validators agree on the next ledger without mining.

Counterparty

Whoever is on the other side of a transaction.

Credential

An on-ledger proof that an account passed a check (XLS-70).

Currency code

A token's name, 3 characters or 40 hex. Anyone can use any code, so it proves nothing alone.

Deep freeze

A freeze where the holder can neither send nor receive the token (XLS-77).

delivered_amount

The amount a payment actually delivered, in its metadata. The figure to credit.

Depth

How much is on offer in an order book.

Destination tag

A number telling a shared account which customer a payment is for.

DEX

The decentralised exchange built into the XRPL.

Digest (SHA-256)

A fixed-length fingerprint of data. Change anything and the digest changes completely.

Domain

A web domain an account claims, verifiable against a file on that domain.

Drops

The smallest unit of XRP: 1 XRP = 1,000,000 drops.

Escrow

Funds locked until a time passes or a condition is met.

Exception

An approved, recorded override of a NO-GO for a stated reason.

Fee pressure

How high transaction costs are running because the network is busy.

Flag

An on/off setting on an account or trust line.

Four-eyes

A control where a second person must approve a change.

Freeze

An issuer power to stop a holder sending its token on.

Funded depth

Depth backed by owners who can actually deliver.

gateway_balances

The server call that returns an issuer's obligations.

Global freeze

An issuer freezing every holder of its tokens at once.

GO

A verdict meaning every check passed.

Hash

A transaction's or object's 64-character hexadecimal identifier.

Hash-chained log

A log where each entry includes the fingerprint of the previous one, so edits show.

HHI

Herfindahl-Hirschman Index: the sum of squared percentage shares; 10,000 is total concentration.

HOLD

A verdict meaning a person should review before moving.

HUD

NOSHASHI's small live panel in the macOS menu bar.

INSUFFICIENT_DATA

A check state meaning the ledger did not give enough to decide. Never a pass.

Investigation

A case in NOSHASHI collecting verdicts, readings and notes, with a hash-chained log.

IOU

Another name for an issued token: a promise from its issuer.

Issuer

The account that creates a token and sets its rules.

KYC

Know Your Customer: verifying who a customer is.

Ledger

One numbered snapshot of the whole XRPL, produced every 3 to 5 seconds.

Ledger index

A ledger's number. Fixes a reading to a moment anyone can re-check.

Limit

The most of a token you are willing to hold on a trust line.

Listed depth

All the depth shown on a book, including offers nobody can fill.

LP token

A token received for adding assets to an AMM pool; carries a fee vote.

Mainnet

The real, live XRP Ledger, as opposed to test networks.

Master key

An account's original key pair. It can be disabled.

Metadata

The record of what a transaction actually did, attached once it is validated.

MiCA

The EU's regulation for crypto-assets, including stablecoin issuers.

Mid price

The average of the best bid and best ask.

Multi-signature

Several keys signing together, set with a signer list.

NFT

A unique token (XLS-20). Its ID encodes the rights its issuer kept.

NFTokenID

The 64-character identifier of an NFT, which encodes its flags.

No Freeze

A permanent promise by an issuer never to freeze.

NO-GO

A verdict meaning a blocking check failed.

NOT_APPLICABLE

A check state meaning the check does not apply.

Obligations

The total of an issuer's tokens held by others: its real circulating supply.

Observer

NOSHASHI's agent mode that watches for changes and opens investigations.

Offer

An order posted on the DEX to trade one asset for another.

Order book

All the offers for one trading pair.

Owner reserve

XRP locked for each object an account owns, such as a trust line or offer.

Partial payment

A payment whose Amount is only a maximum; it can deliver less and still succeed.

Passport

NOSHASHI's signed, portable record of an asset's posture.

Path

A route that lets a payment convert currencies on the way.

Payment channel

Many small off-ledger payments settled later in one transaction.

Permissioned domain

An area where only accounts with the right credentials may take part (XLS-80).

Policy

An organisation's own thresholds and rules applied to verdicts.

Provenance

Where an account came from: its age and first funder.

Quorum

The minimum total signing weight needed to act for an account.

Receipt

NOSHASHI's record of a verdict: ledger index, rules version, checks and digest.

Regular key

A second key an account nominates to sign for it.

Require Auth

An issuer setting: you need its approval to hold its token.

Reserve

XRP an account must keep locked: a base reserve plus an owner reserve per object.

REVIEW

A check state meaning a human decision is needed.

Selective disclosure

Proving a credential exists without revealing the personal data behind it.

Sequence

An account's transaction counter, starting at its creation ledger index.

Signer list

The keys and weights used for multi-signature.

Spread

The gap between the best bid and the best ask.

Stablecoin

A token meant to hold a steady value, such as one US dollar.

taker_gets_funded

The field book_offers adds when an owner cannot cover an offer.

tec

A result prefix: failed, but recorded and the fee charged.

tesSUCCESS

The success result. It does not mean the full Amount arrived.

Token

An asset on the XRPL other than XRP, created by an issuer.

Transaction

A signed instruction that changes the ledger.

Transaction cost

The small XRP fee each transaction pays, which is destroyed.

Transfer fee

A percentage (0% to 100%) an issuer charges when holders send its token to each other.

Travel Rule

FATF Recommendation 16: sender and receiver details travel with transfers above a threshold.

Trust line

The link you open to an issuer to hold its token.

Unfunded offer

An offer still listed although its owner cannot fill it.

UNL

Unique Node List: the validators a server trusts not to collude.

Validated ledger

A ledger at least 80% of trusted validators agreed on. It is final.

Validator

A server that votes on which ledger is the true next one.

Warning check

A check whose failure gives HOLD rather than NO-GO.

Webhook

A signed notification NOSHASHI sends to your own system when an event happens.

X-address

An address format that packs an address and destination tag together.

XLS

A numbered proposal for a new XRPL standard, such as XLS-30 for AMMs.

XRP

The XRP Ledger's own currency.

XRPL

The XRP Ledger.

Under the hood

FOR THE CURIOUS

▼ How NOSHASHI is built

The desktop app uses Tauri 2: a Rust core with a React and TypeScript interface, packaged natively for macOS, Windows and Linux, with an iOS build in progress. It connects to public XRPL servers directly over WebSocket and compares four of them. Accounts, plans and organisation features (policies, exceptions, investigations, webhooks, the Compliance API) run on Supabase edge functions with row-level security. This website is a static site on Vercel with a few server functions, such as the free certificate. Releases are built by GitHub Actions and signed, and installed copies update themselves.

▼ The ledger calls it uses

- `ledger` : the latest validated ledger index.
- `account_info` : an account's balance, flags, sequence and regular key.
- `account_objects` : signer lists, escrows, offers and other owned objects.
- `account_lines` : trust lines, limits, balances and freeze state.
- `gateway_balances` : an issuer's obligations per currency.
- `book_offers` : order books, including `taker_gets_funded` .
- `amm_info` : pool balances, fee, votes and auction slot.
- `tx` and `account_tx` : transactions and their metadata, including `delivered_amount` .

▼ How the authority verdict is computed

Six checks: freeze permanently surrendered (warning), issuance not globally frozen now (blocking), holding does not require issuer permission (warning), no single key controls the issuer (blocking), no issuer transfer fee (warning), and supply concentration (warning; an abstention when the supply was not walked). Any failed blocking check gives NO-GO; otherwise any failed warning gives HOLD; otherwise GO. The certificate records the ledger index, the rules version and a SHA-256 digest over the canonical checks.

▼ Release history

0.1 to 0.3: first public builds. 1.0.4 to 1.0.7: the gate, receipts, policy and investigations. 1.0.8: the Observer, one-click issuer investigation, five-state checks, evidence requests and the live garden. 1.0.9: screens that clipped now scroll, and groundwork for iOS. Full notes are in the [progress log](#).

English ▼